

Applicability of this chapter to configurations:

This chapter is applicable to all CV32A6 configurations.

Configuration	Implementation
CV32A60AX	Implemented extension
CV32A60X	Implemented extension
CV64A6_MMU	Implemented extension

Note: CV64A6 implements RV64M that includes additional instructions.

RV32M Multiplication and Division Instructions

This chapter describes the standard integer multiplication and division instruction extension, which is named “M” and contains instructions that multiply or divide values held in two integer registers.

Multiplication Operations

- **MUL:** Multiplication

Format: mul rd, rs1, rs2

Description: performs a 32-bit × 32-bit multiplication and places the lower 32 bits in the destination register (Both rs1 and rs2 treated as signed numbers).

Pseudocode: $x[rd] = x[rs1] * x[rs2]$

Invalid values: NONE

Exception raised: NONE

- **MULH:** Multiplication Higher

Format: mulh rd, rs1, rs2

Description: performs a 32-bit × 32-bit multiplication and places the upper 32 bits in the destination register of the 64-bit product (Both rs1 and rs2 treated as signed numbers).

Pseudocode: $x[rd] = (x[rs1] \text{ s*s } x[rs2]) \gg_s 32$

Invalid values: NONE

Exception raised: NONE

- **MULHU:** Multiplication Higher Unsigned

Format: mulhu rd, rs1, rs2

Description: performs a 32-bit × 32-bit multiplication and places the upper 32 bits in the destination register of the 64-bit product (Both rs1 and rs2 treated as unsigned numbers).

Pseudocode: $x[rd] = (x[rs1] \text{ u*u } x[rs2]) \gg_u 32$

Invalid values: NONE

Exception raised: NONE

- **MULHSU:** Multiplication Higher Signed Unsigned

Format: mulhsu rd, rs1, rs2

Description: performs a 32-bit × 32-bit multiplication and places the upper 32 bits in the destination register of the 64-bit product (rs1 treated as signed number, rs2 treated as unsigned number).

Pseudocode: $x[rd] = (x[rs1] \text{ s*u } x[rs2]) \gg_s 32$

Invalid values: NONE

Exception raised: NONE

Division Operations

- **DIV:** Division

Format: div rd, rs1, rs2

Description: perform signed integer division of 32 bits by 32 bits (rounding towards zero).

Pseudocode: $x[rd] = x[rs1] /s x[rs2]$

Invalid values: NONE

Exception raised: NONE

- **DIVU:** Division Unsigned

Format: divu rd, rs1, rs2

Description: perform unsigned integer division of 32 bits by 32 bits (rounding towards zero).

Pseudocode: $x[rd] = x[rs1] /u x[rs2]$

Invalid values: NONE

Exception raised: NONE

- **REM:** Remain

Format: rem rd, rs1, rs2

Description: provide the remainder of the corresponding division operation DIV (the sign of rd equals the sign of rs1).

Pseudocode: $x[rd] = x[rs1] \%s x[rs2]$

Invalid values: NONE

Exception raised: NONE

- **REMU:** Remain Unsigned

Format: rem rd, rs1, rs2

Description: provide the remainder of the corresponding division operation DIVU.

Pseudocode: $x[rd] = x[rs1] \%u x[rs2]$

Invalid values: NONE

Exception raised: NONE

